

Homework 6: Metals

Paper track. No home lab (the metals kit stays at practice). Calculator needed.

Part A: Identify the metal (2 pts each)

#	Color	Magnet	Density (g/cm ³)	With HCl	Metal + symbol + atomic #
1	reddish-orange	no	8.96	no reaction	
2	silvery	no	2.70	slow start, then steady	
3	gray	yes	7.87	slow-moderate, solution goes pale green	
4	dull silver	no	1.74	vigorous, fast	
5	bluish-gray	no	7.14	moderate, steady	

Part B: Density (3 pts each)

1. A metal cube 3.0 cm on a side weighs 72.9 g. Density? Which metal?
2. A piece of metal raises water in a graduated cylinder from 25.0 mL to 34.0 mL and weighs 80.6 g. Density? Which metal?
3. Convert 7.14 g/cm³ to kg/m³.
4. A solid iron bar has a volume of 50 cm³. What is its mass? (Fe = 7.87 g/cm³)
5. A 2.0-meter cube of aluminum — what is its mass in kg? (Al = 2700 kg/m³)

Part C: Reactions & oxides (2 pts each)

1. Classify: $2 \text{H}_2\text{O}_2 \rightarrow 2 \text{H}_2\text{O} + \text{O}_2$
2. Classify: $\text{Zn} + 2 \text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$
3. Classify: $2 \text{Mg} + \text{O}_2 \rightarrow 2 \text{MgO}$
4. Why does aluminum resist acid at first, but iron just keeps corroding?
5. Steel, brass, and bronze — name the metals in each.

Part D: Scenario (3 pts)

A gray metal fragment at a scene is not attracted to a magnet, has a measured density of 2.71 g/cm³, and reacts slowly with HCl at first and then steadily. Name the metal. A suspect who works with aircraft parts and a suspect who works with cast-iron cookware are both of interest — which does this implicate?

